Module -4

NOISE POLLUTION AND ITS CONTROL

What is noise?

In simple terms, noise is unwanted sound. Sound is a form of energy which is emitted by a vibrating body and on reaching the ear causes the sensation of hearing through nerves. Sounds produced by all vibrating bodies are not audible. The frequency limits of audibility are from 20 HZ to 20,000 HZ.

A noise problem generally consists of three inter-related elements- the source, the receiver and the transmission path. This transmission path is usually the atmosphere through which the sound is propagated, but can include the structural materials of any building containing the receiver .

Noise may be continuous or intermittent. Noise may be of high frequency or of low frequency which is undesired for a normal hearing. For example, the typical cry of a child roduces sound, which is mostly unfavorable to normal hearing. Since it is unwanted sound, we call it noise. The discrimination and differentiation between sound and noise also depends upon the habit and interest of the person/species receiving it, the ambient conditions and impact of the sound generated during that particular duration of time. There could be instances that, excellently rendered musical concert for example, may be felt as noise and exceptional music as well during the course of the concert!

Sounds of frequencies lessthan 20HZ are called infrasonics and greater than 20,0000HZ are called ultrasonics. Since noise is also a sound, the terms noise and sound are synonymously used and are followed in this module.

Causes, effects and control of noise and thermal pollution.

Noise pollution

Noise is perhaps one of the most undesirable by products of modern mechanized lifestyle. It may not seem as insidious or harmful as the contamination of drinking water supplies from hazardous chemicals, but it is a problem that affects human health and well-being and that can also contribute to the general deterioration of environmental quality. It can affect people at home, in their community, or at their place of work.

Sound waves cause eardrums to vibrate, activating middle and inner organs and sending bioelectrical signals to the brain. The human ear can detect sounds in the frequency range of about 20 to 20,000 Hz, but for most people hearing is best in the range of 200 to 10,000

Hz. A sound of 50 Hz frequency, for example, is perceived to be very low-pitched, and a 15,000 - Hz sound is very high pitched/

Simply defined, noise is undesirable and unwanted sound. It takes energy to produce sound, so, in a manner of speaking, noise is a form of waste energy. It is not a substance that can accumulate in the environment, like most other pollutants, but it can be diluted with distance from a source. All sounds come from a sound source, whether it be a radio, a machine, a human voice, an airplane, or a musical instrument. Not all sound is noise. What may be considered music to one person may be nothing but noise to another. To a extent, noise pollution is a matter of opinion. Noise is measured in terms of Decibel units.

Sources of noise

Based on the type of noise include

a) Industrial Noise

b) Transport Noise

C) Neighbourhood Noise

Measurement of noise

The noise is usually measured either by i) Sound Pressure or ii) Sound Intensity. The

Soun

d intensity is measured in Decibel (dB), which is tenth part of the longest unit "Bel" named

after Alexander Graham Bell. Decibel (dB) is a ratio expressed as the logarithmic scale relative

to a reference sound pressure level. The db is thus expressed as

Intensity Measured (I)

Sound Intensity Level = 10 log ---------------------------

Reference intensity (I_0)

or dB = 10 log I/I_0

Intensity of Noise sources

Sources	Intensity(dB)
Breathing	10
Trickling clock	20-30
Normal conversation	35-60
Office noise	60 - 80
Traffic	50-90
Motor cycle	105
Jet fly	100 -110

Control of Noise Pollution

Noise definitely affects the quality of life. It is therefore important to ensure the mitigation or control of noise pollution. Noise pollution can be controlled

• At source level – Can be done by i) Designing and fabricating silencing devices in air craft engines, automobiles industrial machines and home appliances, ii) By segregating the noisy machines

• During Transmission – can be achieved by adding insulation and sound-proofing to doors, around industrial machinery. Zoning urban areas to maintain a separation between residential areas and zones of excessive noise.Sound

a) Acoustillite : made up of Compressed wood pulp, wood fibers and is available in the form oftiles

b) Acoustical blanket : Prepared from mineral wool or glass fibres

c) Hair Felt: Consists of wool fibres, Coarse Cotton Fibres.

d) Fibre Glass

e) Cork Carpet: Prepared out of pieces of corks treated with linseed oil and is used for covering floors.

f) Acoustic Plaster: Mainly consists of gypsum in the form of plaster.

Protecting the exposed person

• By creating vegetation cover – Plants absorb and dissipate sound energy and thus act as Buffer Zone.

Trees should be planted along highways, schools and other places.

Impacts of noise

Annoyance: It creates annoyance to the receptors due to sound level fluctuations. The aperiodic sound due to its irregular occurrences causes displeasure to hearing and causes annoyance.

Physiological effects: The physiological features like breathing amplitude, blood pressure, heart-beat rate, pulse rate, blood cholesterol are effected.

Loss of hearing: Long exposure to high sound levels cause loss of hearing. This is mostly unnoticed, but has an adverse impact on hearing function.

Human performance: The working performance of workers/human will be affected as they'll be losing theirconcentration.

Nervous system: It causes pain, ringing in the ears, feeling of tiredness, thereby effecting the functioning of human system.

Sleeplessness: It affects the sleeping there by inducing the people to become restless and loose concentration and presence of mind during their activities

Damage to material : The buildings and materials may get damaged by exposure to infrasonic / ultrasonic waves and even get collapsed.

Control of Noise Pollution

Noise generation is associated with most of our daily activities. A healthy human ear responds to a very wide range of SPL from - the threshold of hearing at zero dB, uncomfortable at 100-120 dB and painful at 130-140 dB. Due to the various adverse impacts of noise on humans and environment , noise should be controlled. The technique or the combination of techniques to be employed for noise control depend upon the extent of the noise reduction required, nature of the equipment used and the economy aspects of the available techniques.

The various steps involved in the noise management strategy is illustrated at Reduction in the noise exposure time or isolation of species from the sources form part of the noise control techniques besides providing personal ear protection, engineered control for noise reduction at source and/or diversion in the trajectory of sound waves.

The techniques employed for noise control can be broadly classified as

- ❖ Control at source
- ❖ Control in the transmission path
- ❖ Using protective equipment.

Noise Control at Source

The noise pollution can be controlled at the source of generation itself by employing techniques like-

Reducing the noise levels from domestic sectors: The domestic noise coming from radio, tape recorders, television sets, mixers, washing machines, cooking operations can be minimisedby their selective and judicious operation. By usage of carpets or any absorbing material, the noise generated from felling of items in house can be minimised.

Maintenance of automobiles: Regular servicing and tuning of vehicles will reduce the noise levels. Fixing of silencers to automobiles, two wheelers etc., will reduce the noise levels.

Control over vibrations: The vibrations of materials may be controlled using proper foundations, rubber padding etc. to reduce the noise levels caused by vibrations.

Low voice speaking: Speaking at low voices enough for communication reduces the excess noise levels.

Prohibition on usage of loud speakers: By not permitting the usage of loudspeakers in the habitant zones except for important meetings / functions. Now-a-days, the urban Administration of the metro cities in India, is becoming stringent on usage of loudspeakers.

Selection of machinery: Optimum selection of machinery tools or equipment reduces excess noise levels. For example selection of chairs, or selection of certain machinery/equipment which generate less noise (Sound) due to its superior technology etc. is also an important factor in noise minimization strategy.

Maintenance of machines: Proper lubrication and maintenance of machines, vehicles etc. will reduce noise levels. For example, it is a common experience that, many parts of a vehicle will become loose while on a rugged path of journey. If these loose parts are not properly fitted, they will generate noise and cause annoyance to the driver/passenger. Similarly is the case of machines. Proper handling and regular maintenance is essential not only for noise control but also to improve the life of machine.

Municipal solid waste

Municipal solid waste

Municipal Solid Waste (MSW)—more commonly known as trash or garbage—consists of everyday items we use and then throw away, such as product packaging, grass clippings, furniture, clothing, bottles, food scraps, newspapers, appliances, paint, and batteries. This comes from our homes, schools, hospitals, and businesses.

Types of Municipal Solid Wastes

Municipal solid waste can be divided into the following categories given below:

1. **Kitchen Waste:** This waste contains fruits and vegetable peels. It is biodegradable and can be used to increase the fertility of the soil.
2. **Office or School Waste:** This waste contains paper, cardboard, plastic clips, crayons, markers etc. these are usually recyclable and non-biodegradable.
3. **Hazardous Waste:** this waste is collected from hospitals, clinics, or medical centres. They contain chemicals and already used syringes, pieces of cotton, creams, or ointments. These must be disposed of very carefully using special treatments to avoid infections or diseases.

Types of Solid Waste Management

Landfill: It involves burying the waste in vacant locations around the city. The dumping site should be covered with soil to prevent contamination.
Benefits: A sanitary disposal method if managed effectively.
Limitations: A reasonably large area is required.

Incineration: It is the controlled oxidation (burning/thermal treatment) of mostly organic compounds at high temperatures to produce thermal energy, CO_2, and water.
Benefits: Burning significantly reduces the volume of combustible waste.
Limitations: Smoke and fire hazards may exist.

Composting: It is a natural process of recycling organic matter like leaves and food scraps into beneficial fertilizers that can benefit both soil and plants.
Benefits: It is beneficial for crops and is an environment-friendly method.
Limitations: Requires high-skilled labour for large-scale operation.

Recycling: It is a process of converting waste material into new material. Examples: wood recycling, paper recycling, and glass recycling.
Benefits: It is environment-friendly.
Limitations: It is expensive to set up and not reliable in case of an emergency.

Vermicomposting: Vermicomposting is a bio-conversion technique that is commonly used to handle solid waste. Earthworms feed on organic waste to reproduce and multiply in number, vermicompost, and vermi wash as products in this bio-conversion process.
Benefits: It reduces the need for chemical fertilizers and enhances plant growth.
Limitations: It is time-consuming, cost-ineffective, and requires extra care.

Various Methods of Solid Waste Management
Municipal Solid Waste

Every day goods such as product packaging, yard trimmings, furniture, clothing, bottles, cans, food, newspapers, appliances, electronics, and batteries make up the municipal solid waste.
With rising urbanisation and change in lifestyle, the amount of municipal waste is also rising.

It is roughly classified into five categories:

1. Recyclable Material: Glasses, bottles, cans, paper, metals, etc.
2. Composite Wastes: Tetra packs, toys.
3. Biodegradable Wastes: Kitchen waste, flowers, vegetables, fruits, and leaves.
4. Inert Waste: Rocks, debris, construction material.
5. Domestic Hazardous and Toxic Waste: E-waste, medication, light bulbs, etc.

Harmful Effects of Solid Waste

- Bad odour of waste
- Production of toxic gases
- Degradation of natural beauty
- Air pollution
- Water pollution
- Soil pollution
- Spread of diseases
- Effect on biodiversity

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Characteristics of solid waste

Physical characteristics

Information and data on the physical characteristics of solid wastes are important for the selection and operation of equipment and for the analysis and design of disposal facilities. The following physical characteristics are to be studied in detail.

Density

Density of waste, i.e., its mass per unit volume (kg/m3), is a critical factor in the design of a solid waste management system, e.g., the design of sanitary landfills, storage, types of collection and transport vehicles, etc. To explain, an efficient operation of a landfill demands compaction of wastes to optimum density. Any normal compaction equipment can achieve reduction in volume of wastes by 75%, which increases an initial density of 100 kg/m3 to 400 kg/m3. In other words, a waste collection vehicle can haul four times the weight of waste in its compacted state than when it is uncompacted. Significant changes in density occur spontaneously as the waste moves from source to disposal, due to scavenging, handling, wetting and drying by the weather, vibration in the collection vehicle and decomposition.

Moisture content

Moisture content is defined as the ratio of the weight of water (wet weight - dry weight) to the total wet weight of the waste. Moisture increases the weight of solid wastes, and thereby, the cost of collection and transport. In addition, moisture content is a critical determinant in the economic feasibility of waste treatment by incineration, because wet waste consumes energy for evaporation of water and in raising the temperature of water vapour. In the main, wastes should be insulated from rainfall or other extraneous water.

A typical range of moisture content is 20 to 40%, representing the extremes of wastes in an arid climate and in the wet season of a region of high precipitation. However, values greater than 40% are not uncommon. Climatic conditions apart, moisture content is genera generally higher in low income countries because of the higher proportion of food and yard waste.

Size of Waste constituents

The size distribution of waste constituents in the waste stream is important because of its significance in the design of mechanical separators and shredder and waste treatment process. This varies widely and while designing a system, proper analysis of the waste characteristics should be carried out.

Calorific Value

Calorific value is the amount of heat generated from combustion of a unit weight of a substance, expressed as kcal/kg. The calorific value is determined experimentally using Bomb calorimeter in which the heat generated at a constant temperature of 25OC from the combustion of a dry sample is measured.

The physical properties that are essential to analyse of wastes disposed at landfills are:

Field capacity

The field capacity of municipal solid waste is the total amount of moisture which can be retained in a waste sample subject to gravitational pull. It is a critical measure because water in excess of field capacity will form leachate, and leachate can be a major problem in landfills. Field capacity varies with the degree of applied pressure and the state of decomposition of the wastes.

Permeability of compacted wastes

The hydraulic conductivity of compacted wastes is an important physical property because it governs the movement of liquids and gases in a landfill. Permeability depends on the other properties of the solid material include pore size distribution, surface area and porosity. Porosity represents the amount of voids per unit total volume of material. The porosity of municipal solid waste varies typically from 0.40 to 0.67 depending on the compaction and composition of the waste.

Compressibility

It is the degree of physical changes of the suspended solids or filter cake when subjected to pressure.

Chemical characteristics

Knowledge of the classification of chemical compounds and their characteristics is essential for the proper understanding of the behaviour of waste, as it moves through the waste management system. The products of decomposition and heating values are two examples of chemical characteristics. If solid wastes are to be used as fuel, or are used for any other purpose, we must know their chemical characteristics, including the following

Chemical: Chemical characteristics include pH, Nitrogen, Phosphorus and Potassium (N-P-K), total Carbon, C/N ratio, calorific value.

Bio-Chemical: Bio-Chemical characteristics include carbohydrates, proteins, natural fibre, and biodegradable factor.

Toxic: Toxicity characteristics include heavy metals, pesticides, insecticides, Toxicity test for Leachates (TCLP), etc.

Lipids

This class of compounds includes fats, oils and grease. Lipids have high calorific values, about 38000 kcal/kg, which makes waste with a high lipid content suitable for energy recovery processes. Since lipids in the solid state become liquid at temperatures slightly above ambient, they add to the liquid content during waste decomposition. They are biodegradable but because they have a low solubility in waste, the rate of biodegradation is relatively slow.

Carbohydrates

Carbohydrates are found primarily in food and yard waste. They include sugars and polymers of sugars such as starch and cellulose and have the general formula (CH2O)X. Carbohydrates are readily biodegraded to products such as carbon dioxide, water and methane. Decomposing carbohydrates are particularly attractive for flies and rats and for this reason should not be left exposed for periods longer than is necessary.

Proteins

Proteins are compounds containing carbon, hydrogen, oxygen and nitrogen and consist of an organic acid with a substituted amine group (NH2). They are found mainly in food and garden wastes and comprise 5-10% of the dry solids in solid waste. Proteins decompose to form amino acids but partial decomposition can result in the production of amines, which have intensely unpleasant odours.

Natural fibres

This class includes the natural compounds, cellulose and lignin, both of which are resistant to biodegradation. They are found in paper and paper products and in food and yard waste. Cellulose is a larger polymer of glucose while lignin is composed of a group of monomers of which benzene is the primary member. Paper, cotton and wood products are 100%, 95% and 40% cellulose respectively. Since they are highly combustible, solid waste having a high proportion of paper and wood products, are suitable for incineration. The calorific values of ovendried paper products are in the range 12000 – 18000 kcal/kg and of wood about 20000 kcal/kg, which compare with 44200 kcal/kg for fuel oil.

Synthetic organic material (Plastics)

They are highly resistant to biodegradation and, therefore, are objectionable and of special concern in solid waste management. Hence the increasing attention being paid to the recycling of plastics to reduce the proportion of this waste component at disposal sites. Plastics have a high heating value, about 32,000 kJ/kg, which make them very suitable for incineration. But, one should note that polyvinyl chloride (PVC), when burnt, produces dioxin and acid gas. The latter increases corrosion in the combustion system and is responsible for acid rain.

Functional Elements of Solid Waste Management System comprises of six basic elements including:

1. Generation of the solid waste
2. On-site handling & storage
3. Collection
4. Transfer & transport
5. Material and resource recovery and
6. Disposal

Generation of Solid Waste

Solid waste generation refers to the creation of waste materials by residential, commercial, and industrial activities. This function involves understanding the sources, quantities, and composition of waste to develop appropriate waste management strategies. Waste generation rates can be estimated using the following equation:

Waste Generation = Population × Per Capita Waste Generation Rate

For example, if a city has a population of 100,000 and the per capita waste generation rate is 1.2 kg/person/day, the estimated waste generation would be 120,000 kg/day.

The handling, storage, and separation of solid waste at the source before they are collected is a critical step in the management of residential solid waste

Waste Handling

Handling refers to activities associated with managing solid wastes until they are placed in the containers used for their storage before collection or return to drop-off and recycling centers.

On-site handling and storage involve the proper containment and temporary storage of waste at the point of generation. It includes activities such as waste segregation, waste minimization, and the use of appropriate containers or bins. For instance, households may use separate bins for recyclable materials (e.g., paper, plastic, glass) and non-recyclable waste. On-site storage ensures that waste is properly managed until it is collected.

Waste Storage

The first phase to manage solid waste is at the home level. It requires temporary storage of refuse on the premises. The individual household or businessman has responsibility for the onsite storage of solid waste. For individual homes, industries, and other commercial centers, proper on-site storage of solid waste is the beginning of proper disposal, because unkept solid waste or simple dumps are sources of nuisance, flies, smells, and other hazards.

Collection

The collection is the process of gathering and transporting waste from various sources to a central location or transfer station. Efficient collection systems are designed based on factors such as population density, waste generation rates, and transportation logistics. Collection methods include curbside collection, door-to-door collection, and community drop-off points. Collection vehicles, such as garbage trucks, are used to transport waste to the next stage.

Transfer and Transport

Transfer and transport involve the movement of waste from the collection points or transfer stations to treatment or disposal facilities. Waste transfer stations act as intermediate hubs where waste is consolidated from smaller collection vehicles into larger transport vehicles. Proper handling, containment, and transport equipment are essential to prevent spillage, littering, and odor nuisances during transfer and transport processes.

Resource Recovery and Processing

Examples of material and resource recovery include:

Recycling: Separating and processing recyclable materials such as paper, plastic, metal, and glass for remanufacturing into new products.

Composting: Decomposing organic waste materials, such as food scraps and yard waste, to produce nutrient-rich compost for soil amendment.

Energy Recovery: Utilizing waste as a fuel source through processes like waste-to-energy (WTE) or anaerobic digestion to generate electricity or heat.

Disposal

Disposal is the final stage of solid waste management when waste that cannot be recovered or recycled is safely and responsibly disposed of. Common disposal methods include landfilling and incineration.

Landfilling: Waste is placed in a specially engineered landfill, where it undergoes controlled decomposition over time. Landfills are designed with liners, leachate collection systems, and gas management infrastructure to minimize environmental impacts.

Incineration: Waste is burned at high temperatures in waste-to-energy facilities, reducing its volume and generating electricity or heat. Modern incineration plants are equipped with air pollution control technologies to minimize emissions.

People's Participation in solid waste management

Role of people is Essential in the Following Areas

1. Reduce, Reuse & Recycling (R R R) of waste.
2. Not to throw the waste/litter on the streets, drains, open spaces, water bodies, etc.
3. Storage of organic/bio-degradable and recyclable waste separately at source.
4. Primary collection of waste
5. Community storage/collection of waste in flats, multi-storied buildings, societies, commercial complexes, etc.
6. Managing excreta of pet dogs and cats appropriately.
7. Waste processing/disposal at a community level (optional)
8. Pay adequately for the services provided.